
Linear Methods and Regularization

Lecture 2 ■ POLSCI 689L

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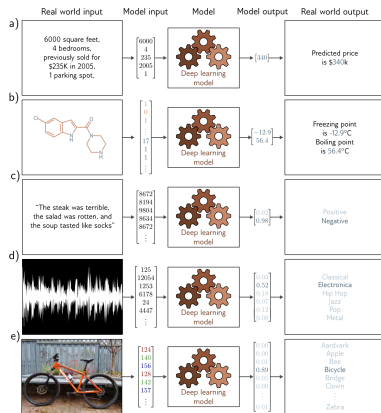
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Overview

1. **Learning problems:** supervised and unsupervised learning
2. **Generalization:** training error, test error, and the bias–variance trade-off
3. **Linear models:** low-dimensional and high-dimensional settings
4. **LASSO:** sparsity, tuning, prediction rates, inference, and support recovery
5. **Ridge, elastic net, and Lava:** dense and sparse-plus-dense coefficient structures

Supervised learning observes inputs and outcomes

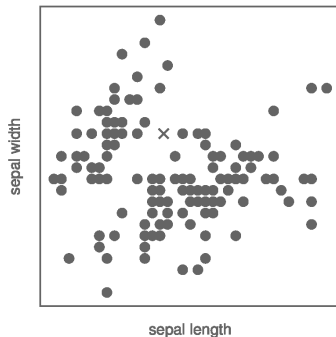
- We observe training examples (X_i, Y_i) .
- We estimate $\hat{f}$ from those labeled examples.
- We use $\hat{f}(X_{\text{new}})$ to predict the outcome for a new input.



Source: [Prince, 2023].

Unsupervised learning has no outcome labels

- We observe X but not a response Y .
- The goal is to uncover or describe the underlying structure of the data.
- Examples include identifying topics in documents and grouping documents by their content.



Key idea

Supervised learning predicts a labeled outcome; unsupervised learning searches for structure without labels.

Statistical learning estimates an unknown function

Suppose we observe a response Y and p predictors $X = (X_1, \dots, X_p)$. A general model is

$$Y = f(X) + \epsilon, \quad f(X) = \mathbb{E}[Y \mid X].$$

- The function f captures the systematic relationship between X and Y .
- The error ϵ captures variation not explained by X .
- Statistical learning consists of methods for estimating the unknown function f .

Key idea

The learning problem begins with an unknown function and data that provide information about it.

Training error measures in-sample fit

Given the observed sample, a natural criterion is the sample mean squared error:

$$\text{MSE}_{\text{train}} = \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}(x_i) \right)^2.$$

- We use the training sample to fit $\hat{f}$, so training error can be very small.
- A sufficiently flexible method may fit not only systematic structure but also sample-specific noise.
- The method with the lowest training MSE need not have the lowest test MSE.

Key idea

Good in-sample fit is not the same as good prediction on unseen data.

Test error has two components

For a new test point (x_0, y_0) , the out-of-sample MSE is

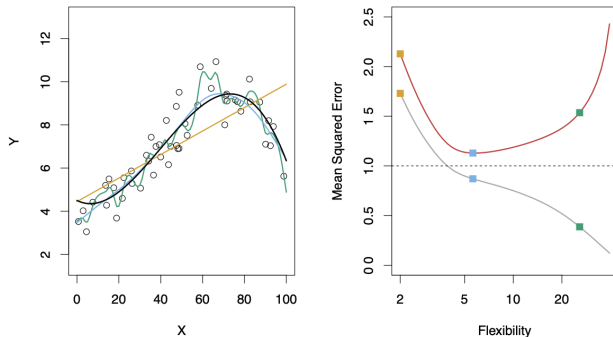
$$\mathbb{E}\left[\left(y_0 - \hat{f}(x_0)\right)^2\right].$$

Recalling that $y = f(x) + \epsilon$,

$$\mathbb{E}\left[\left(y_0 - \hat{f}(x_0)\right)^2\right] = \mathbb{E}\left[\left(f(x_0) - \hat{f}(x_0)\right)^2\right] + \text{Var}(\epsilon).$$

- The first term is *reducible error*: an appropriate learning method may lower it.
- The second term is *irreducible error*: it remains even if f is estimated perfectly.

Model flexibility can create underfitting or overfitting



Training MSE is shown in gray; test MSE is shown in red.

Key idea

Increasing flexibility lowers training error, but test error may eventually rise.

The bias–variance decomposition

For a fixed test input x_0 ,

$$\begin{aligned}\mathbb{E}\left[\left(y_0 - \hat{f}(x_0)\right)^2\right] &= \underbrace{\left(f(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2}_{\text{Bias}^2(\hat{f}(x_0))} \\ &\quad + \underbrace{\mathbb{E}\left[\left(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2\right]}_{\text{Var}(\hat{f}(x_0))} + \underbrace{\text{Var}(\epsilon)}_{\text{irreducible error}}.\end{aligned}$$

Key idea

A good learning method must control both systematic approximation error and sensitivity to the training sample.

Flexibility trades bias for variance

Variance

Variance measures how much $\hat{f}$ would change if it were estimated using a different training set. More flexible methods generally have higher variance.

Bias

Bias is the error introduced by approximating a complicated relationship with a simpler model. More flexible methods generally have lower bias.

Key idea

As flexibility increases, variance tends to rise while bias tends to fall. When a given method yields a small training MSE but a large test MSE, we are said to be overfitting the data.

Training, validation, and test sets

Training set

Estimate model parameters for each candidate specification.

Validation set

Choose tuning parameters and compare candidate models.

Test set

Evaluate the final procedure after model selection is complete.

Key idea

Separating model fitting, model selection, and final evaluation protects the assessment of generalization.

K-fold cross-validation reuses scarce training data

1. Split the training data into K non-overlapping folds.
2. Fit the model on $K - 1$ folds and validate it on the remaining fold.
3. Repeat the procedure K times so that each fold serves once as validation data.
4. Average the validation errors across the K experiments.
 - Common choices are $K = 5$ or $K = 10$.
 - Cross-validation estimates generalization error without permanently setting aside a large validation sample.

The linear model in low dimensions

Given an i.i.d. sample $D_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, consider

$$Y_i = X_i^T \beta_0 + \epsilon_i \quad \Longleftrightarrow \quad Y = X \beta_0 + \epsilon,$$

where $Y \in \mathbb{R}^n$ and $X \in \mathbb{R}^{n \times p}$. The OLS estimator is

$$\hat{\beta}^{\text{OLS}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - X_i^T \beta)^2 = (X^T X)^{-1} X^T Y.$$

If X and ϵ are independent and $\epsilon \sim N(0, \sigma^2 I)$, then

$$\mathbb{E} \left[\frac{1}{n} \|X \hat{\beta}^{\text{OLS}} - X \beta_0\|_2^2 \right] = \sigma^2 \frac{p}{n}.$$

When $p \ll n$, this in-sample prediction risk is small.

High dimensions challenge ordinary least squares

- When p/n is not small, the risk $\sigma^2 p/n$ is no longer negligible.
- When $p > n$, $X^T X$ is singular and the least-squares coefficient vector is not unique.
- With more parameters than observations, a linear model may interpolate the training data.
- Zero training error can coexist with large generalization error.

Key idea

The response is to restrict model complexity through regularization.

Regularization deliberately constrains the fit

A regularized linear estimator solves a problem of the form

$$\hat{\beta} = \arg \min_{\beta} \{ \text{RSS}(\beta) + \lambda P(\beta) \}.$$

- The penalty $P(\beta)$ restricts the effective complexity of the fitted model.
- The tuning parameter λ controls the amount of shrinkage.
- Regularization introduces bias in exchange for lower variance.

Ridge regression uses an ℓ_2 penalty

Ridge regression shrinks coefficients by constraining their ℓ_2 norm:

$$\begin{aligned}\hat{\beta}^{\text{ridge}} &= \arg \min_{\beta} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 \\ &\text{subject to } \sum_{j=1}^p \beta_j^2 \leq t \quad (\|\beta\|_2^2 \leq t).\end{aligned}$$

The equivalent penalized form is

$$\hat{\beta}^{\text{ridge}} = \arg \min_{\beta} \left\{ \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}.$$

As λ increases, ridge estimates shrink toward zero; as λ decreases, they approach the OLS estimates.

Ridge regression has a unique closed-form solution

In matrix form, the objective is

$$(\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta) + \lambda\beta^T\beta.$$

For any $\lambda > 0$, it is strictly convex and has the unique solution

$$\hat{\beta}^{\text{ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y}.$$

- Adding λ to the diagonal makes the problem nonsingular even when $\mathbf{X}^T\mathbf{X}$ is not full rank.
- Ridge is not equivariant to the scaling of the inputs, so predictors are normally standardized first.
- If $\mathbf{X}^T\mathbf{X} = \mathbf{I}$, then $\hat{\beta}^{\text{ridge}} = \hat{\beta}^{\text{OLS}}/(1 + \lambda)$.
- Ridge is biased but has lower variance than OLS.

LASSO uses an ℓ_1 penalty

LASSO shrinks regression coefficients by constraining their ℓ_1 norm:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2$$

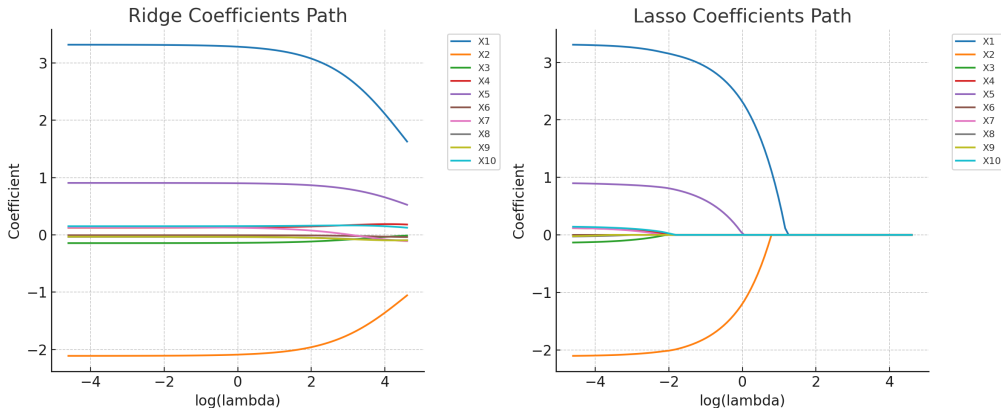
subject to $\sum_{j=1}^p |\beta_j| \leq t \quad (\|\beta\|_1 \leq t).$

An equivalent penalized form is

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \left\{ \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}.$$

The tuning parameter λ controls the amount of shrinkage.

Coefficient paths show how shrinkage changes the fit



Coefficient paths for ridge regression and LASSO.

KKT conditions characterize the LASSO solution

For the objective $\frac{1}{2}\|Y - X\beta\|_2^2 + \lambda\|\beta\|_1$, the Karush–Kuhn–Tucker condition is

$$X^T(Y - X\hat{\beta}) = \lambda s,$$

where s is a subgradient of the ℓ_1 norm evaluated at $\hat{\beta}$:

$$s_j \in \begin{cases} +1, & \hat{\beta}_j > 0, \\ -1, & \hat{\beta}_j < 0, \\ [-1, 1], & \hat{\beta}_j = 0. \end{cases}$$

- The fitted value $X\hat{\beta}$ is unique because squared-error loss is strictly convex in $X\beta$.
- The coefficient vector $\hat{\beta}$ need not be unique when $X^T X$ is singular, as can occur when $p > n$.
- Even when LASSO coefficients are not unique, any two solutions agree on the signs of their common nonzero coefficients.

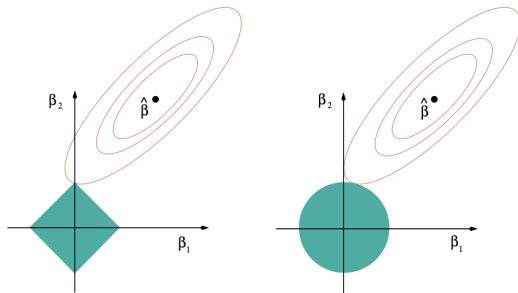
The active-set solution reveals shrinkage

Let $A = \text{supp}(\hat{\beta})$ denote the active set. Then

$$\begin{aligned}\hat{\beta}_A &= (X_A^T X_A)^{-1} (X_A^T y - \lambda s_A), \\ \hat{\beta}_{-A} &= 0.\end{aligned}$$

- On the selected support, LASSO shrinks the OLS estimate by $\lambda(X_A^T X_A)^{-1} s_A$.

The ℓ_1 constraint can produce sparse solutions



The ellipses are contours of the objective function centered at the OLS estimates; the green regions are the constraints.

Key idea

The corners of the ℓ_1 constraint allow LASSO solutions to have coefficients exactly equal to zero.

Average loss and standardized predictors

A common implementation minimizes average squared error:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \left\{ \frac{1}{2N} \sum_{i=1}^N \left(y_i - \beta_0 - \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}.$$

- This scaling makes λ independent of the sample size.
- Predictors are typically standardized to have mean zero and variance one, especially when they are measured in different units.
- The corresponding active-set expression is $\hat{\beta}_A = (X_A^T X_A)^{-1} (X_A^T y - N \lambda s_A)$.

K-fold Cross-validation chooses the LASSO penalty

1. Randomly divide the full dataset into $K > 1$ groups; common choices are $K = 5$ or $K = 10$.
2. Use one group as validation data and the remaining $K - 1$ groups as training data.
3. Fit LASSO for a range of λ values and record validation MSE.
4. Repeat until every group has served as validation data, then average the K errors for each λ .

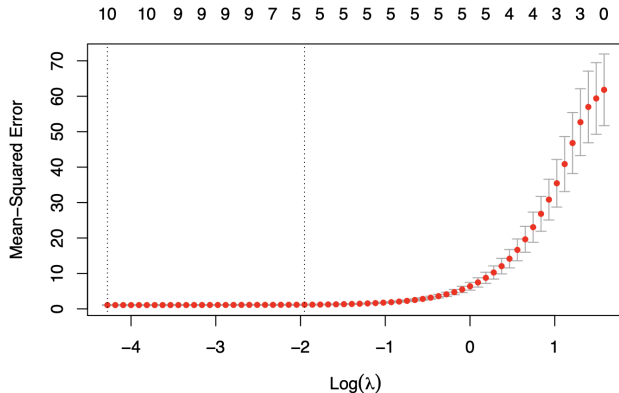
$\lambda_{\min}$

The value of λ with the lowest cross-validation error. It typically yields a more complex model with lower bias and potentially higher variance.

λ_{1se}

The largest λ whose error is within one standard error of the minimum. It favors a simpler model with somewhat higher bias and lower variance.

The cross-validation curve guides model selection



Source: [Hastie et al., 2015].

Statistical inference after LASSO is difficult

- The distribution of the LASSO estimator is difficult to characterize.
- Because of the penalty, some coefficients are set exactly to zero with positive probability, even asymptotically.
- The asymptotic distribution is irregular and depends on the tuning parameters and the true sparsity pattern.
- LASSO also creates a post-selection inference problem because the model is chosen through a data-driven process, whereas classical statistical theory typically assumes a fixed model.

Key idea

Strong prediction performance does not automatically provide valid coefficient-level inference.

A slow-rate result bounds LASSO prediction error

We ask how close $X\hat{\beta}$ is to $X\beta_0$. For any $\delta > 0$, a slow-rate bound is

$$\frac{1}{n} \|X\hat{\beta} - X\beta_0\|_2^2 \leq 4\sigma \|\beta_0\|_1 \sqrt{\frac{2 \log(ep/\delta)}{n}}$$

with probability at least $1 - \delta$.

- This is called a *slow-rate* result because it is of order $\sqrt{\log(p)/n}$.
- For comparison, the in-sample risk of low-dimensional linear regression is $\sigma^2 p/n$.
- The rate will matter later when LASSO is used inside double machine learning.

Proof step 1: begin with the basic inequality

Because $\hat{\beta}$ minimizes the objective,

$$\frac{1}{2}\|Y - X\hat{\beta}\|_2^2 + \lambda\|\hat{\beta}\|_1 \leq \frac{1}{2}\|Y - X\beta_0\|_2^2 + \lambda\|\beta_0\|_1.$$

Using $Y = X\beta_0 + \epsilon$ and rearranging,

$$\begin{aligned}\|X\hat{\beta} - X\beta_0\|_2^2 &\leq 2\langle \epsilon, X\hat{\beta} - X\beta_0 \rangle \\ &\quad + 2\lambda \left(\|\beta_0\|_1 - \|\hat{\beta}\|_1 \right).\end{aligned}$$

Here $\langle a, b \rangle = a^T b$.

Proof step 2: control the stochastic term

Hölder's inequality gives

$$\langle \epsilon, X\hat{\beta} - X\beta_0 \rangle \leq \|X^T \epsilon\|_\infty \|\hat{\beta} - \beta_0\|_1.$$

By the triangle inequality,

$$\begin{aligned} \|X\hat{\beta} - X\beta_0\|_2^2 &\leq 2\|X^T \epsilon\|_\infty \left(\|\hat{\beta}\|_1 + \|\beta_0\|_1 \right) \\ &\quad + 2\lambda \left(\|\beta_0\|_1 - \|\hat{\beta}\|_1 \right). \end{aligned}$$

If $\lambda \geq \|X^T \epsilon\|_\infty$, then

$$\|X\hat{\beta} - X\beta_0\|_2^2 \leq 4\lambda \|\beta_0\|_1.$$

Proof step 3: choose λ

Suppose $\max_j \|X_j\|_2 \leq \sqrt{n}$ and $\epsilon \sim N(0, \sigma^2 I)$. Then, for any $\delta > 0$,

$$\mathbb{P}\left(\|X^T \epsilon\|_\infty \leq \sigma \sqrt{2n \log(ep/\delta)}\right) \geq 1 - \delta.$$

Taking

$$\lambda = \sigma \sqrt{2n \log(ep/\delta)}$$

and dividing the previous bound by n yields

$$\frac{1}{n} \|X\hat{\beta} - X\beta_0\|_2^2 \leq 4\sigma \|\beta_0\|_1 \sqrt{\frac{2 \log(ep/\delta)}{n}}$$

with probability at least $1 - \delta$.

Variable selection

- Prediction analysis studies ℓ_2 error or the prediction error of a LASSO solution.
- Variable-selection analysis asks whether LASSO recovers the true sparsity pattern: whether $\hat{\beta}$ is nonzero in the same positions as β^* .
- This property is called variable-selection consistency, or *sparsistency*.
- The error $\|\hat{\beta} - \beta^*\|_2$ can be small even when $\hat{\beta}$ and β^* have different supports.

Conditions for consistent variable selection

Irrepresentability / mutual incoherence

There must be some $\gamma > 0$ such that

$$\max_{j \in S^c} \left\| (X_S^T X_S)^{-1} X_S^T x_j \right\|_1 \leq 1 - \gamma.$$

This is a form of near orthogonality between relevant and irrelevant predictors.

Eigenvalue condition

The active submatrix must satisfy

$$\lambda_{\min}(X_S^T X_S / N) \geq C_{\min} > 0.$$

Otherwise, the active predictors are linearly dependent and β^* cannot be estimated even if S were known.

Variable-selection theorem: assumptions

Suppose that:

1. the design matrix X satisfies mutual incoherence with parameter $\gamma > 0$;
2. the columns of X are normalized;
3. the eigenvalue condition holds with constant $C_{\min} > 0$; and
4. the noise vector e has i.i.d. $N(0, \sigma^2)$ entries.

For the regularized LASSO program, choose

$$\lambda_N \geq \frac{8\sigma}{\gamma} \sqrt{\frac{\log p}{N}}.$$

Then, with probability greater than

$$1 - c_1 \exp(-c_2 N \lambda_N^2),$$

the unique optimal solution satisfies the following support-recovery properties.

Variable-selection theorem: conclusions

No false inclusion

$$S(\hat{\beta}) \subseteq S(\beta^*).$$

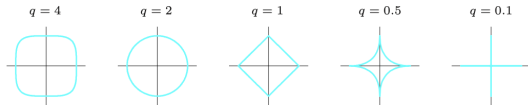
No false exclusion above the beta-min threshold

The solution includes every $j \in S(\beta^*)$ for which

$$|\beta_j^*| > \beta_{\min}, \quad \beta_{\min} = \lambda_N \left[\frac{4\sigma}{\sqrt{C_{\min}}} + \left\| (X_S^T X_S / N)^{-1} \right\|_{\infty} \right].$$

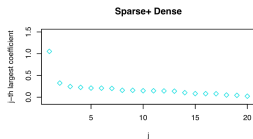
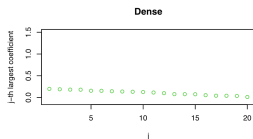
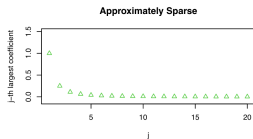
LASSO, ridge, and coefficient structure

- LASSO is well suited to approximately sparse settings.
- Ridge penalizes large coefficients more aggressively and small coefficients less aggressively, making it well suited to dense models.
- More generally, one can consider an ℓ_q penalty $\sum_{j=1}^p |\beta_j|^q$.



Source: [Hastie et al., 2009].

Coefficient patterns can be sparse, dense, or both



- **Approximately sparse:** a few coefficients are large, while the remaining coefficients decay rapidly toward zero.
- **Dense:** many coefficients are small but nonzero.
- **Sparse plus dense:** several coefficients are large, while many small coefficients do not decay rapidly enough for approximate sparsity.

LASSO is designed for approximate sparsity, ridge for dense signals, and Lava for a sparse-plus-dense structure.

Elastic net combines the LASSO and ridge penalties

Elastic-net regularization solves

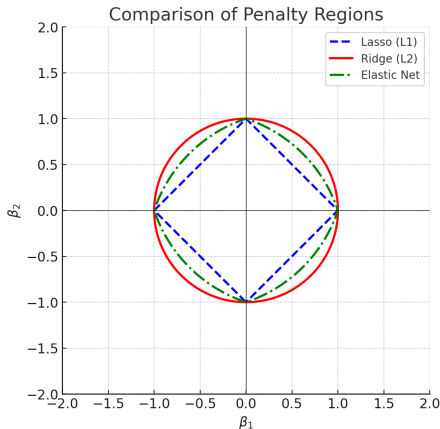
$$\hat{\beta}^{\text{EN}} = \arg \min_{\beta} \left\{ \frac{1}{N} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \right\}.$$

An equivalent parameterization writes the penalty as

$$\lambda \sum_{j=1}^p \left\{ \alpha \beta_j^2 + (1 - \alpha) |\beta_j| \right\}, \quad 0 \leq \alpha \leq 1.$$

- Under this parameterization, $\alpha = 0$ gives LASSO and $\alpha = 1$ gives ridge.
- Elastic net was developed to reduce LASSO's high variability when predictors are highly correlated.
- With appropriate tuning, it can perform well in either sparse or dense settings.

Elastic net combines two penalty geometries



- The ℓ_1 constraint has corners and can produce exact zeros.
- The ℓ_2 constraint is smooth and shrinks correlated coefficients together.
- The elastic-net region lies between these two geometries.

Key idea

Elastic net subjects the same coefficient vector to both penalties.

Lava decomposes the coefficient vector

Lava represents the coefficient vector as $\beta = \delta + \xi$, where δ is dense and ξ is sparse. It solves

$$(\hat{\delta}, \hat{\xi}) = \arg \min_{\delta, \xi} \left\{ \frac{1}{N} \|\mathbf{y} - \mathbf{X}(\delta + \xi)\|_2^2 + \lambda_2 \|\delta\|_2^2 + \lambda_1 \|\xi\|_1 \right\}, \quad \hat{\beta}^{\text{Lava}} = \hat{\delta} + \hat{\xi}.$$

- The ridge penalty handles the dense component δ ; the LASSO penalty handles the sparse component ξ .
- The resulting penalty behaves like ridge for small coefficients and like LASSO for large coefficients.
- Like ridge, Lava generally produces a dense estimate and therefore does not perform variable selection.

Key idea

Elastic net mixes two penalties on one vector; Lava assigns them to separate components.

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